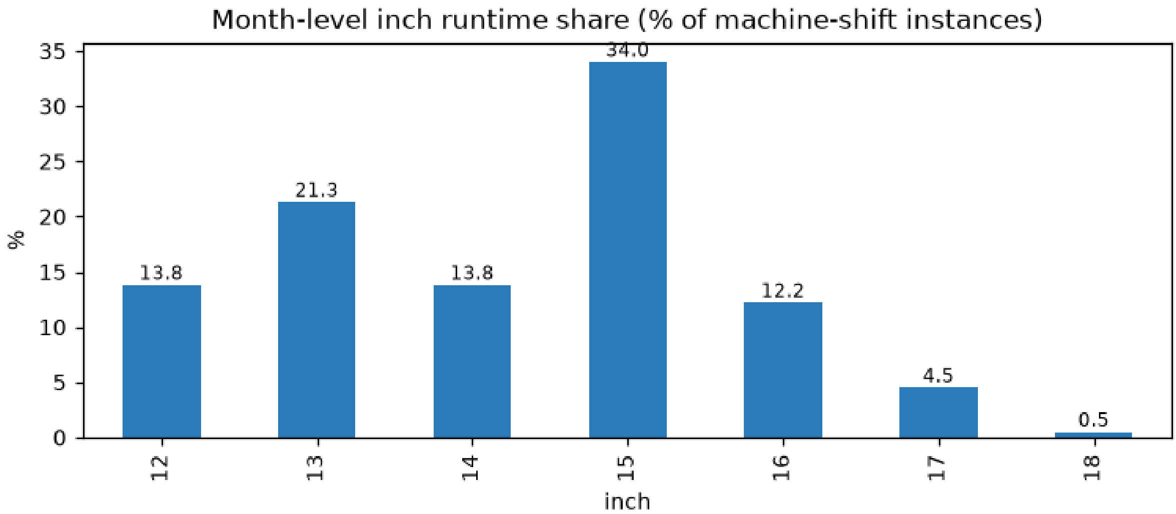
# Inch-Run Study — May Plant Production

*Machine / Group / Inch behaviour analysis & requirements for a perfect building schedule*

Prepared from actual May production (machine × day × shift × inch, 30 building machines, 32 days, 4 MG groups).

## 1. Executive summary

- **Inch demand is concentrated:** 15" (34%), 13" (21%), 14" (14%), 12" (14%), 16" (12%) make up ~95% of run-time; 17"/18" are tails.
- **The plant is mostly — but not strictly — inch-locked:** 66% of machine-days run a single inch; the rest run a dominant inch plus minor others.
- **Each MG group has a distinct inch policy** (UNISTAGE strict 12/13; BJ on 13/15/16; TBM on 12/15; VMIMAXX flexible 14–18).
- **Every machine has ONE dominant inch** carrying the bulk of its runtime — the practical 'inch-lock' — with a small allowed set around it.
- **Implication:** a perfect building schedule needs the *real per-machine allowed-inch set + dominant inch*, not a single hard inch — see Section 6.

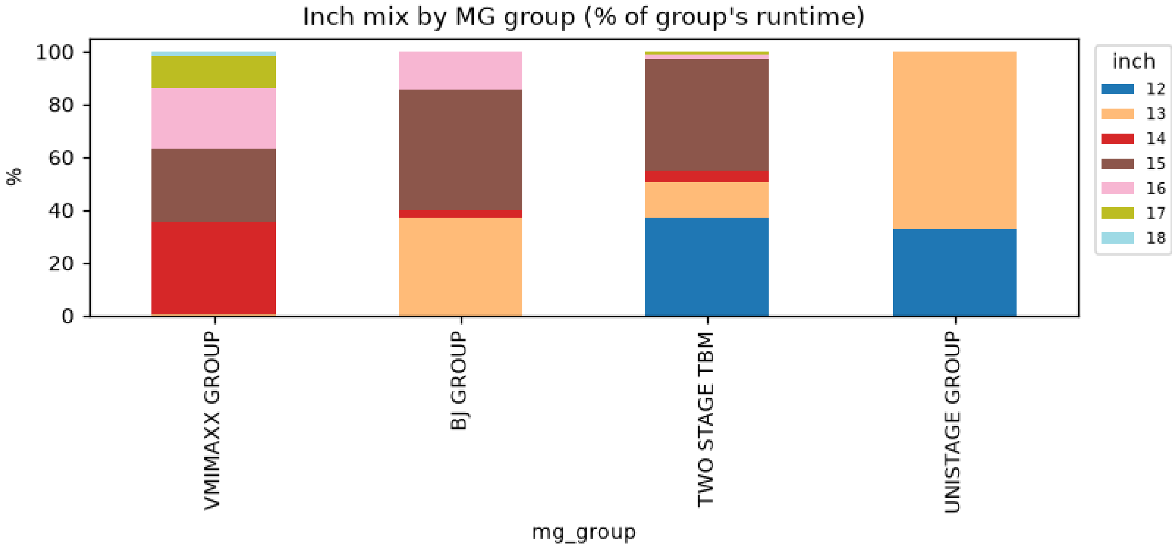


## 2. Group-level inch policy

Each group behaves differently — the scheduler must respect these as group rules:

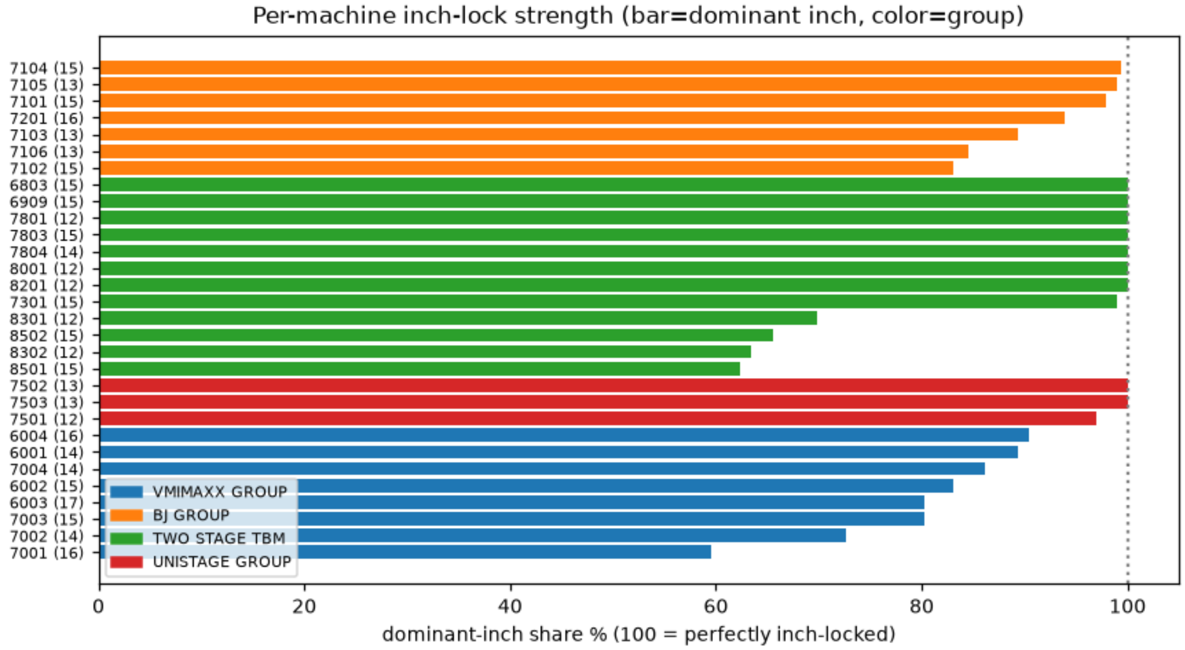
- **UNISTAGE** — perfectly inch-locked: only 12" (33%) and 13" (67%). Strictest group.
- **BJ GROUP** — 13" (37%), 15" (45%), 16" (15%); well-locked per machine (83–99% dominant).
- **TWO-STAGE TBM** — 12" (37%), 15" (43%), 13" (13%); ~half its machines single-inch.
- **VMIMAXX** — the flexible group: 14" (35%), 15" (28%), 16" (23%), 17" (12%); machines carry 4–7 inches. Absorbs the varied/high-inch work.

| Group | 12" | 13" | 14" | 15" | 16" | 17" | 18" |
|---|---|---|---|---|---|---|---|
| VMIMAXX | 0 | 0 | 35 | 28 | 23 | 12 | 2 |
| BJ | 0 | 37 | 2 | 45 | 15 | 0 | 0 |
| TWO STAGE TBM | 37 | 13 | 4 | 43 | 1 | 1 | 0 |
| UNISTAGE | 33 | 67 | 0 | 0 | 0 | 0 | 0 |



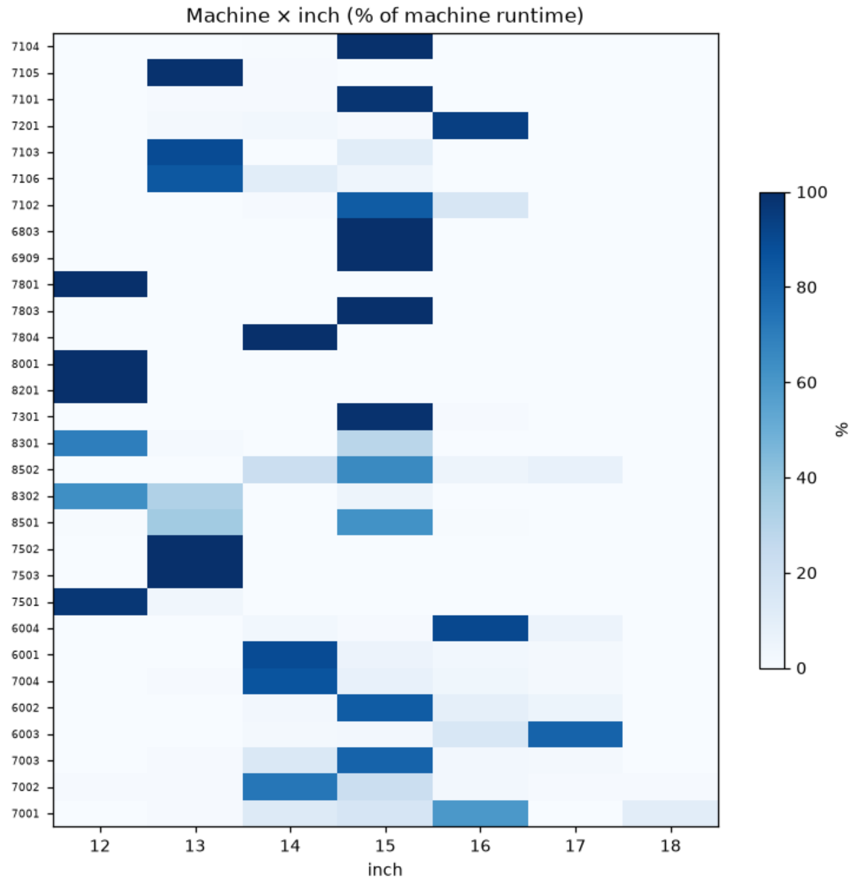
## 3. Machine-level inch-lock strength

Dominant-inch share = how strongly each machine sticks to one inch (100% = perfectly locked). UNISTAGE & many TBM machines are 100%; VMIMAXX machines are the loosest (60–90%, up to 7 inches).



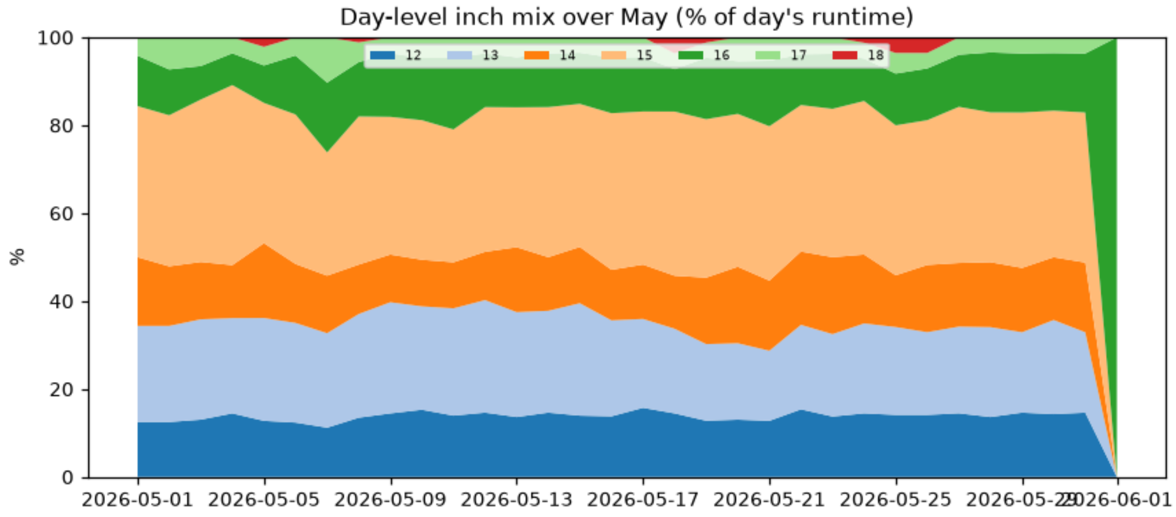

| Machine | Group | Dom. inch | Dom. share | #inches |
|---|---|---|---|---|
| 7104 | BJ | 15" | 99% | 2 |
| 7105 | BJ | 13" | 99% | 2 |
| 7101 | BJ | 15" | 98% | 3 |
| 7201 | BJ | 16" | 94% | 4 |
| 7103 | BJ | 13" | 89% | 2 |
| 7106 | BJ | 13" | 84% | 3 |
| 7102 | BJ | 15" | 83% | 3 |
| 6803 | TWO STAGE TBM | 15" | 100% | 1 |
| 6909 | TWO STAGE TBM | 15" | 100% | 1 |
| 7801 | TWO STAGE TBM | 12" | 100% | 1 |
| 7803 | TWO STAGE TBM | 15" | 100% | 1 |
| 7804 | TWO STAGE TBM | 14" | 100% | 1 |
| 8001 | TWO STAGE TBM | 12" | 100% | 1 |
| 8201 | TWO STAGE TBM | 12" | 100% | 1 |
| 7301 | TWO STAGE TBM | 15" | 99% | 2 |
| 8301 | TWO STAGE TBM | 12" | 70% | 3 |

| Machine | Group | Dom. inch | Dom. share | #inches |
|---|---|---|---|---|
| 8502 | TWO STAGE TBM | 15" | 66% | 4 |
| 8302 | TWO STAGE TBM | 12" | 63% | 3 |
| 8501 | TWO STAGE TBM | 15" | 62% | 4 |
| 7502 | UNISTAGE | 13" | 100% | 1 |
| 7503 | UNISTAGE | 13" | 100% | 1 |
| 7501 | UNISTAGE | 12" | 97% | 2 |
| 6004 | VMIMAXX | 16" | 90% | 4 |
| 6001 | VMIMAXX | 14" | 89% | 4 |
| 7004 | VMIMAXX | 14" | 86% | 5 |
| 6002 | VMIMAXX | 15" | 83% | 4 |
| 6003 | VMIMAXX | 17" | 80% | 4 |
| 7003 | VMIMAXX | 15" | 80% | 5 |
| 7002 | VMIMAXX | 14" | 73% | 7 |
| 7001 | VMIMAXX | 16" | 60% | 5 |



## 4. Day-level inch mix (May)

The daily inch split is fairly stable across the month — there isn't a clean day-to-day rotation; inches run largely in parallel every day, in proportion to demand.




## 5. Validation of the manual analysis

- VMIMAXX 'tries to maintain same inch, sometimes different' — **confirmed**: 59–90% dominant, but 4–7 inches each.
- '1 of 7 inches has more share on each machine' — **confirmed**: every machine has a clear dominant inch.
- '14,15,16,17 frequent' (VMIMAXX) — **confirmed**.
- 'UNISTAGE perfectly inch-locked 12/13' — **confirmed** (100% single-inch machines).

## 6. What we need for a PERFECT building schedule

From a production-planning / scheduling standpoint, the inch study says the current single-hard-inch assumption is wrong. To build the perfect schedule we need the following master data & rules:

| Information / rule needed | Why it matters for the schedule |
|---|---|
| **Per-machine ALLOWED-INCH SET (not one inch)** | Each machine runs a dominant inch + a small allowed set (e.g. VMIMAXX 6001 → 14 primary, also 15/16/18). Capture the real allowed set so the scheduler can use the legitimate flexibility instead of stranding a machine when its single inch is done. |
| **Per-machine DOMINANT / PREFERRED inch** | Use as the soft inch-lock: prefer the dominant inch, allow others only to absorb overflow — minimises size-change changeovers while keeping machines busy. |
| **Group-level inch POLICY** | UNISTAGE = strict (12/13 only); VMIMAXX = flexible (14–18); BJ/TBM = 2–3 inches. Encode per-group allowed inches as a hard constraint. |
| **CHANGEOVER COST matrix (same-size vs size-change)** | Inch (size) change costs far more than a formula change. The schedule must price size-changes so it only switches inch when it genuinely relieves the tail. |
| **Inch-wise DEMAND vs group CAPACITY** | Match inch demand (15">13">14"≈12">16") to the groups/machines that run that inch, by capacity — so no inch is over-routed to a small/slow group. |
| **Per-machine CYCLE TIME by inch** | Cycle time varies by machine and product; needed for accurate capacity & load-levelling per inch. |
| **Real machine↔SKU eligibility** | Routing eligibility should reflect the inch sets above (the tbs2 'current running machine' map is a good seed) rather than a broad routing list. |

**Bottom line:** model inch as a per-machine *allowed-set with a preferred (dominant) inch* and a size-change changeover cost — not a single hard inch. That lets the scheduler keep the strict groups (UNISTAGE) locked, use the flexible group (VMIMAXX) to absorb the tail, and stop stranding machines once their primary inch is exhausted.